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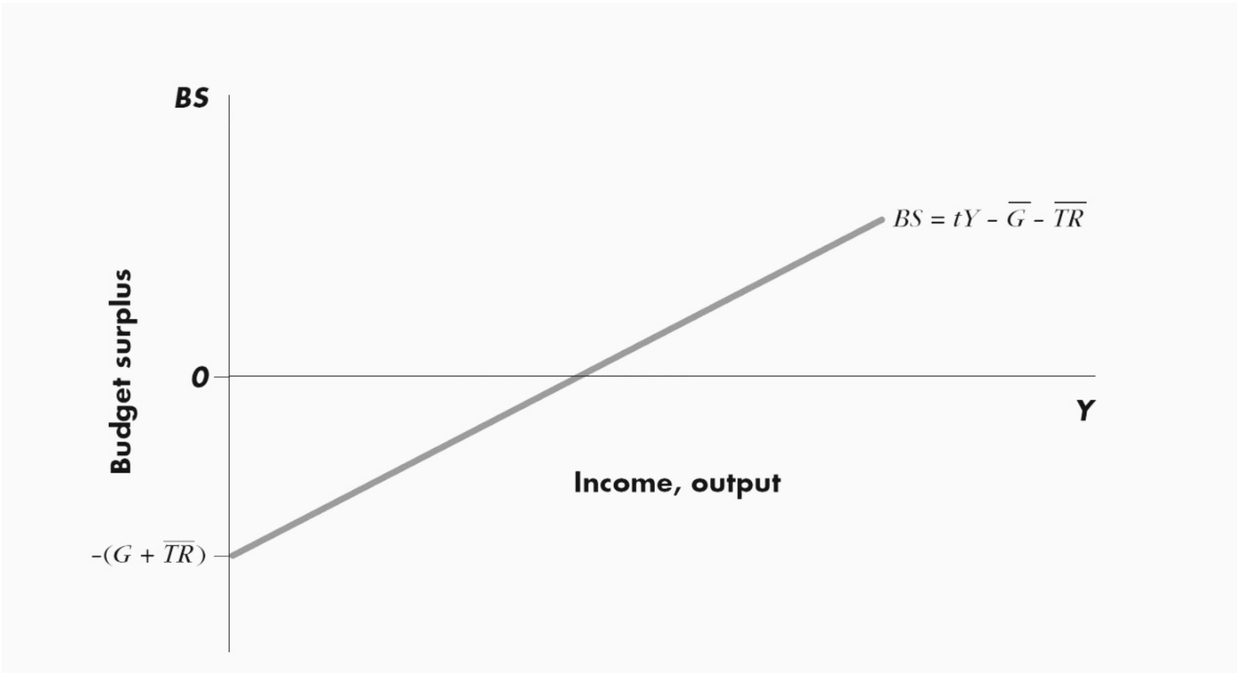
1. Explain the concept of balanced budget multiplier.

Answer: A balanced budget multiplier is to gauge the change in aggregate demand brought about by a change in government expenditure, which is balanced by a change in revenues received from taxes and other sources. That is, the government increases purchases of goods and other commodities, at the same time increases its taxation to keep the budget deficit or surplus unchanged. Because the government then spends the money, spending is increased in the aggregate, which drives economic growth. This multiplier is useful in the analysis of fiscal policy changes that involves both government purchases and taxes.

First there is the budget surplus(BS). It is the excess of Government's revenues from taxes over its total expenditures, consisting of goods and services and transfer payments. Then there is the budget deficit which is the excess of expenditure over revenue (negative surplus).

As a result of the balanced budget multiplier, the BS will stay the same. Only the equilibrium output will increase, according to the government purchases. The balanced-budget multiplier is equal to one. The "positive" impact on aggregate production caused by a change in government purchases is largely, but not completely, offset by the "negative" impact of the change in taxes. The only part of the impact of the change in government purchases NOT offset by the change in taxes is the purchase of aggregate production made by the initial injection. Hence, the change in aggregate production is equal to the initial change in government purchases.

A balanced budget increase in government purchases will increase the equilibrium level of GDP.



An equal increase in autonomous government purchases and autonomous taxes will shift the BS function upwards and lead to an increase in the equilibrium level of GDP. An equal decrease in autonomous government purchases and autonomous taxes will shift the BS function downwards and lead to a decrease in the equilibrium level of GDP

The budget surplus depends not just on Government's favoured policies but also on something that shifts the level of income. Consequently, an increase in government expenditure and taxation of equal amounts will have a net expansionary effect on aggregate demand and incomes, while a decrease in government expenditure and taxation of equal amounts will have a net contractionary effect.

2. How can certain instruments of fiscal policy help in automatically stabilizing the economy through their influence on the simple Keynesian multiplier?

Fiscal policy is the means by which a government adjusts its spending levels and tax rates to monitor and influence a nation's economy. It is the sister strategy to monetary policy through which a central bank influences a nation's money supply. These two policies are used in various combinations to direct a country's economic goals.Based on the theories of British economist John Maynard Keynes, hence also named as keynesian economics.

Governments can influence macroeconomic productivity levels by increasing or decreasing tax levels and expenditure. This influence curbs inflation (generally considered to be healthy when between 2% and 3%), increases employment, and maintains a healthy value of money. Fiscal policy plays a very important role in managing a country's economy. That is, if the economy is producing less than potential output, government spending can be used to employ idle resources and boost output. Increased government spending will result in increased aggregate demand, which then increases the real GDP, resulting in an rise in prices. This is known as expansionary fiscal policy.

Conversely, in times of economic expansion, the government can adopt a contractionary policy, decreasing spending, which decreases aggregate demand and the real GDP, resulting in a decrease in prices. In instances of recession, government spending does not have to make up for the entire output gap. There is a multiplier effect that boosts the impact of government spending. The government could stimulate a great deal of new production with a modest expenditure increase if the people who receive this money consume most of it. This extra spending allows businesses to hire more people and pay them, which in turn allows a further increase in spending, and so on in a virtuous circle.

In addition to changes in spending, the government can also close recessionary gaps by decreasing income taxes, which increases aggregate demand and real GDP, which in turn increases prices. Conversely, to close an expansionary gap, the government would increase income taxes, which decreases aggregate demand, the real GDP, and then prices.

The idea of Keynesian multiplier originated in an explanation of the favourable effects of investment on total employment but it has become part and parcel of the Keynesian theory of income and employment. It is designed to show the relationship of a small increase in investment to cause a much larger increase in income. The multiplier mechanism suggested that heavy spending-by government, business or consumers- would have a salutary impact on the expansion of national income.

It is the ratio of the total change in income to the initial change in investment. In other words, it is the ratio expressing the quantitative relationship between the increase in national income and the increase in investment which induces the rise in income.

3. How does investment sensitivity to interest rate influence the slope of the IS curve? What determines the position of the IS curve?

The Slope and Position of the IS Curve: When Investment is on the x-axis, and the interest on the y-axis, it's negatively sloped. This is because as interest rate gets higher, investment spending reduces the aggregate demand and eqb level of income. Typically the independent variable that is, investment is put on the y-axis, but we'll be using the interest rate as the independent variable,when we graph the IS Graph, for consistency.

The steepness of the curve depends on:(1)- how sensitive investment spending is to changes in the interest rate (2)- on the multiplier (K).

Exogenous investment determines the initial level. Again, a higher interest rate results in lower investment spending. A high interest rate means firms reduce their investment spending to avoid high interest payments. A low interest rate means firms can increase their capital spending and pay relatively low interest. A given change in the interest rate produces a large change in aggregate demand, and thus shifts the aggregate demand curve up by a large distance, if sensitivity is high.

A large shift in the aggregate demands schedule produces a correspondingly large change in the equilibrium level of income. If a given change in the interest rate produces a large change in income, the IS curve is very flat. This is the case if investment is very sensitive to changes in interest rate. On the opposite, if the investment spending is not much sensitive to changes in the interest rate, the IS curve is relatively steep.

Now, we already know that changes in investment spending bring about changes in income depending upon the value of the multiplier. If investment spending is very sensitive to changes in interest rate, the value of multiplier is large and hence the change in income is also large which leads to flattening of the IS curve. If the investment spending is relatively insensitive to changes in the rate of interest, the IS curve is steep because of the lower value of the multiplier.

Given that the slope of the IS curve depends on the multiplier; fiscal policy can affect that slope. The multiplier is affected by the tax rate: an increase in the tax rate reduces the multiplier and that makes the IS curve steeper.

How about the position of the IS curve? How does the IS curve shift? The answer is through an increase in investment spending. If investment expenditure increases, income increases multiplier (K) times the change in investment. This shifts the IS curve horizontally by a distance equal to the multiplier times the change in autonomous spending.

Accordingly, an increase in government purchases or transfer payments will shift the IS curve out to the right, the extent of the shift depending on the size of the multiplier. A reduction in transfer payments or in government purchases shifts the IS curve to the left.

4. When there is reduction in money supply, how does it influence the LM schedule? Derive the aggregate demand schedule using the concept of IS-LM equilibrium.

Suppose we hold the nominal money supply constant. Now if the price level (P) rises, the supply of real money balances (M/P) falls. As a result the LM curve shifts upwards to the left. This leads to a rise in r and a fall in Y as shown in part (a) of Fig. 11.1.

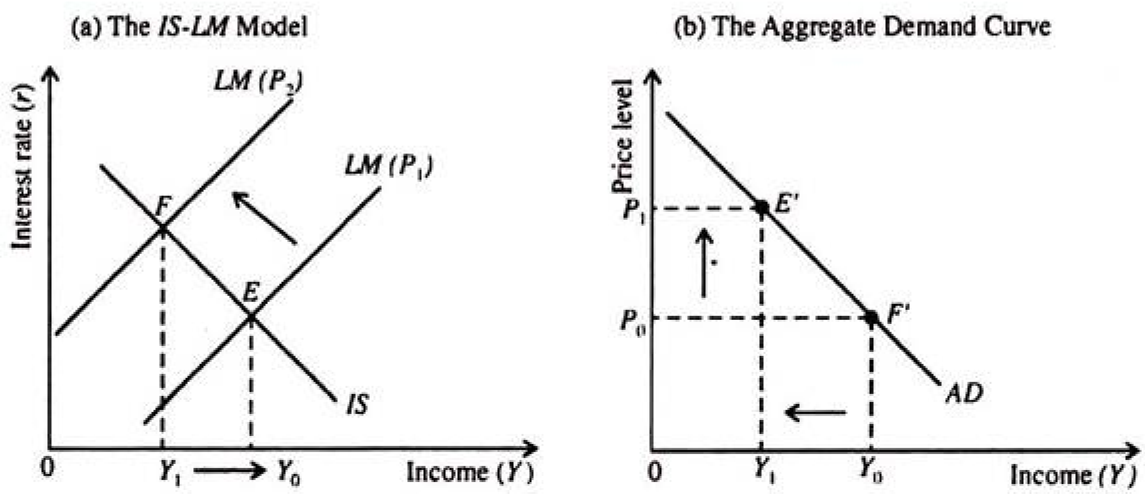


Fig. 11.1 Deriving the Aggregate Demand Curve with the *IS-LM* Model

We see that as the price level rises from P0 to P1 the income level falls to from Y0 to Y1. This inverse relationship between Y and P is captured by the aggregate demand curve, as shown in part (b) of Fig. 11.1.

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Thus the aggregate demand curve is a locus of points showing alternative combinations of P and Y that are consistent with the general equilibrium of the goods market and money market, i.e., equilibrium r and Y — shown by the intersection of the IS and LM curves.

The aggregate demand curve shifts due to any event that shifts the IS curve or the LM curve (when P remains constant). For instance, if M increases Y rises if P remains constant. As a result aggregate demand curve shifts to the right as shown in part (a) of Fig. 11.2. The converse is also true. A fall in M reduces Y and shifts the aggregate demand curve to the left.

Similarly for a constant price level, an increase in G or a cut in T shifts the aggregate demand curve to the right, as shown in part (b) of Fig. 11.2. The converse is also true. A fall in G or an increase in T lowers Y or shifts the aggregate demand curve to the left.

Diagram below:

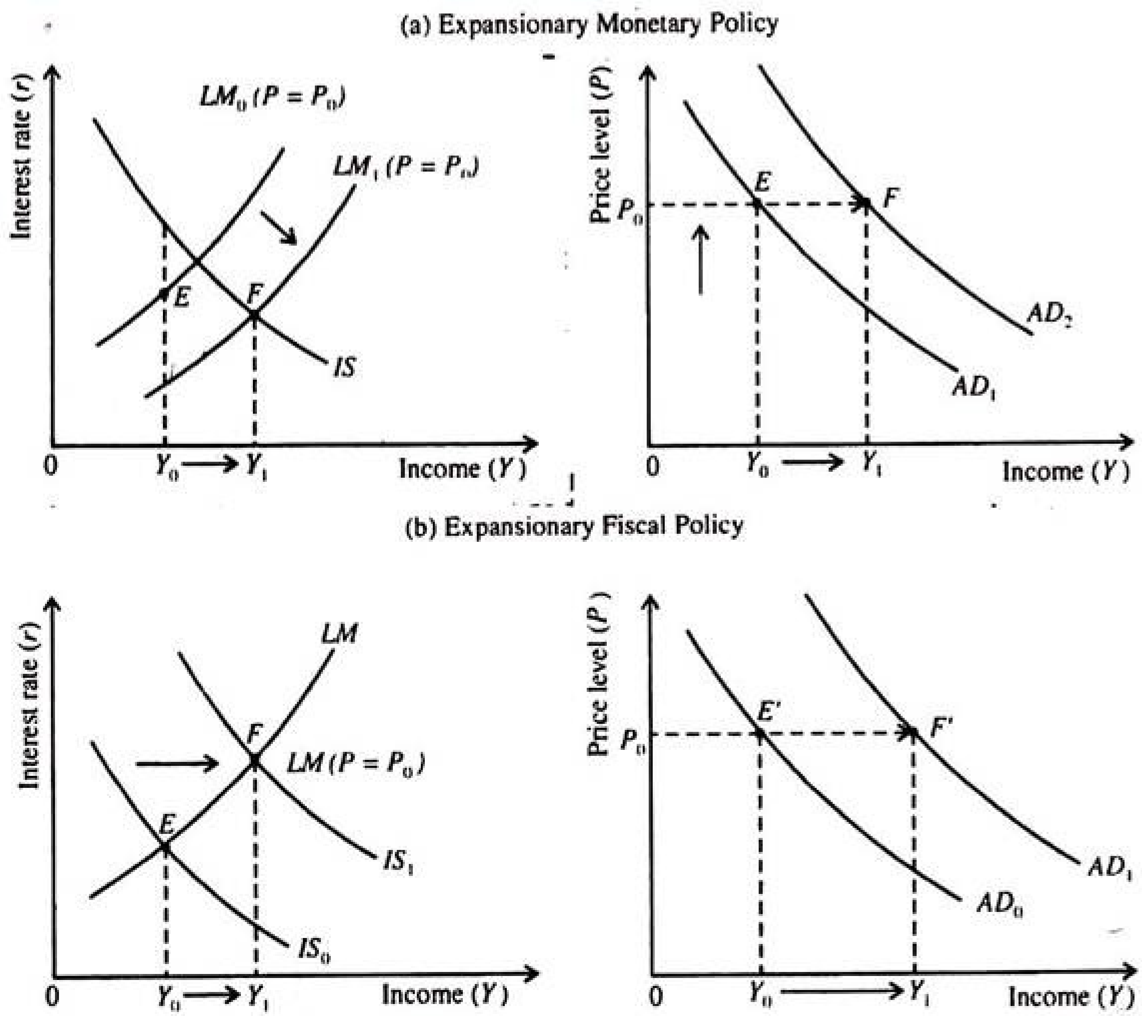

Fig. 11.2 How Monetary and Fiscal Policies Shift the Aggregate Demand Curve